



Number Theory

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Division

Definition:

- $a \mid b$ iff $\exists c: b = ac$
- a divides b , b is a multiple of a , a is a factor of b
- If a does not divide b , we write $a \nmid b$

Properties:

1. $a \mid b, b \mid c \rightarrow a \mid c$
2. $a \mid b, a \mid c \rightarrow a \mid bx + cy$
3. $a \mid b, b \mid a \leftrightarrow |a| = |b|$
4. $a \mid 1 \leftrightarrow |a| = 1$
5. $a \mid b \rightarrow a \mid bc$
6. $\forall n \geq 1: a \mid b \leftrightarrow a^n \mid b^n$

Some Proofs

Proof (2):

$$\begin{aligned} a|b, a|c &\rightarrow \exists b', c': b = ab', c = ac' \rightarrow bx = ab'x, cy = ac'y \\ &\rightarrow bx + cy = a(b'x + c'y) \rightarrow a|bx + cy \end{aligned}$$

Proof (6):

$$a|b \rightarrow \exists c: b = ac \rightarrow b^n = a^n c^n \rightarrow a^n|b^n$$

The reverse is not simple now; wait for GCD.

Some Problems

Problem:

if $x_1x_2 + x_2x_3 + \cdots + x_nx_1 = 0$ and $x_i \in \{1, -1\}$, then $4|n$

Solution:

It is clear $2|n$. So, $n/2$ terms are $+1$ and $n/2$ terms -1

Multiply all terms. In one hand we have $(-1)^{n/2}$. In the other hand we have $(x_1x_2x_3 \dots x_n)^2 = 1$. Then

$$(-1)^{n/2} = 1 \rightarrow 4|n$$

Problem:

$$a - c \mid ab + cd \rightarrow a - c \mid ad + bc$$

Solution:

$$a - c \mid (ad + bc) - (ab + cd) = (a - c)(d - b)$$

Some Problems

Problem: $13|4^{2n+1} + 3^{n+2}$

Solution:

- Basis Step: $13|4^1 + 3^2$
- Inductive Step:

$$13|4^{2n+1} + 3^{n+2} \rightarrow 13|4^{2(n+1)+1} + 3^{(n+1)+2}$$

$$\begin{aligned} & (4^{2(n+1)+1} + 3^{(n+1)+2}) - (4^{2n+1} + 3^{n+2}) \\ &= 15 \times 4^{2n+1} + 2 \times 3^{n+2} = 13 \times 4^{2n+1} + 2 (4^{2n+1} + 3^{n+2}) \end{aligned}$$

Problem: $9|a^2 + ab + b^2 \rightarrow 3|a, 3|b$

Solution:

$$\begin{aligned} 9|a^2 + ab + b^2 &= (a - b)^2 + 3ab \rightarrow 3|a - b \\ &\rightarrow 9|(a - b)^2 \rightarrow 9|3ab \rightarrow 3|a \vee 3|b \\ (3|a - b) \wedge (3|a \vee 3|b) &\rightarrow 3|a \wedge 3|b \end{aligned}$$

Greatest Common Divisor

Definition:

$GCD(a, b) = d$ iff

1. $d \mid a, d \mid b$

2. $\forall d': d' \mid a, d' \mid b \rightarrow d' \leq d$

Properties (Let 's denote $GCD(a, b)$ by (a, b)):

1. $(a, b) = (b, a) = (-a, b) = (a, -b) = (-a, -b)$

2. $\forall k: (a, b) = (a, b + ak)$

3. $\exists x, y: ax + by = (a, b)$

4. $(a, b) = (a, c) = 1 \rightarrow (a, bc) = 1$

5. $(a^n, b^n) = (a, b)^n$

6. $a \mid bc, (a, b) = 1 \rightarrow a \mid c$

7. $(ka, kb) = |k|(a, b)$

8. $a \mid c, b \mid c, (a, b) = 1 \rightarrow ab \mid c$

• < 1. —

قضیه بزرگ

Some Proofs

Proof (2)

Let $(a, b) = d$, $(a, b + ak) = d'$

$(a, b) = d \rightarrow d|a, d|b \rightarrow d|a, d|b + ak \rightarrow d \leq d'$

Similarly $d' \leq d$, and then $d = d'$

$d'|a$
 $d'|b+ak$
 $d'|b$
 $d'|a$
 $d' \leq d$

Proof (6)

$(a, b) = 1 \rightarrow \exists x, y: ax + by = 1 \rightarrow acx + bcy = c$

$\rightarrow acx + aa'y = c \rightarrow a(cx + a'y) = c \rightarrow a|c$

k'

Proof (8)

$(a, b) = 1 \rightarrow \exists x, y: ax + by = 1 \rightarrow acx + bcy = c$

$\rightarrow abb'x + baa'y = c \rightarrow ab(b'x + a'y) = c \rightarrow ab|c$

« قضیه بزرگ » :

برای هر دو عدد صحیح حداقل یکی غیر صفر وجود دارد x, y صحیح بطوریکه

$$(a, b) = ax + by$$

بعضی م
proof:

« بزرگترین مقسوم علیه مشترک a, b برابر است با

کوچکترین ترکیب خطی مثبت a, b »

$$S = \{ax + by \mid x, y \in \mathbb{Z} \text{ and } ax + by > 0\}$$

مجموعه‌ای از اعداد صحیح +

Well-ordering principle (اصل خوش‌ترتیب)

$$d = \min(S)$$

$$\underline{d = ax_0 + by_0}$$

ترکیب خطی از a, b

دش $d \nmid a$

قضیه تقسیم $a = dq + r$

$$0 < r < d$$

$$r = \underline{dq} - a = q(ax_0 + by_0) - a = a(qx_0 - 1) - by_0$$

$$\underline{r \in S}, \underline{r < d}, \underline{d = \min(S)}$$

$\Rightarrow \underline{d \mid a}$ به همین صورت بابر $d \mid b$ اثبات می‌کنیم.

$$\forall d', \begin{cases} d' \mid a \text{ and} \\ d' \mid b \end{cases}$$

$$\begin{aligned} d' \mid ax_0 + by_0 &= d \\ d' \mid d &\quad \underline{d' \leq d} \end{aligned}$$

d بزرگترین است زیرا

d — $\checkmark$

$$(a, b) = (a, c) = 1$$

اثبت 4:

$$ax + by = 1$$

$$au + cv = 1$$

$$\rightarrow (ax + by)(au + cv) = 1$$

$$a(xu + xv + au) + bc(y) = 1$$

عكس قیاسی (عکس قیاسی) $(a', b') = 1 \leftarrow a'x + b'y = 1$

..... $(a, b, c) = 1$

$$(a, b) = d$$

$$d \mid a$$

$$d \mid b$$

$$d \mid ax + by = 1$$

$$d \mid 1 \Rightarrow d = 1$$

اثبت - عکس قیاسی قیاسی

$$(a^n, b^n) = (a, b)^n$$

اثبات 5:

ر.ر. $d = (a, b)$

$$a = md$$

$$b = nd$$

$$\underline{(m, n) = 1}$$

$$a^n = m^n d^n$$

$$b^n = n^n d^n \Rightarrow (a^n, b^n) = d^n$$

Some Problems

Problem: show $(3n + 2, 7n + 5) = 1$

Solution 1:

$$(3n + 2, 7n + 5) = d \rightarrow d \mid 3n + 2, d \mid 7n + 5 \rightarrow \\ d \mid 7(3n + 2) - 3(7n + 5) = -1 \rightarrow d = 1$$

Solution 2:

$$(3n + 2, 7n + 5) = (3n + 2, 7n + 5 - 2(3n + 2)) \\ = (3n + 2, n + 1) = (n + 1, 3n + 2) \\ = (n + 1, 3n + 2 - 3(n + 1)) = (n + 1, -1) = 1$$

Problem: $a \mid b \iff a^n \mid b^n$

Solution: $a^n \mid b^n \rightarrow (a^n, b^n) = a^n \rightarrow (a, b) = a \rightarrow a \mid b$

Some Problems

Definition:

if $(a, b) = 1$, they are called relatively prime

Problem: Among 5 consecutive numbers, there is one which is relatively prime to the other four numbers

Solution:

for any $|a - b| < 5$, we know $(a, b) = 1, 2, 3, \text{ or } 4$

It suffices to show there is a number x s.t. $(x, 6) = 1$ $2 \nmid x$
 $3 \nmid x$

Between 5 consecutive numbers, there are two consecutive odd numbers. One of these two is not divisible by 3; otherwise their difference which is 2 must be divisible by 3. This number is the answer.

Problem: Prove problem for 16 consecutive numbers

Least Common Multiples

Definition:

- $LCM[a, b] = L$ iff
 1. $L > 0$
 2. $a | L, b | L$
 3. $\forall L': a | L', b | L' \rightarrow L \leq L'$

سوکمترین مضرب مشترک

Properties (Let 's denote $LCM(a, b)$ by $[a, b]$):

1. $[a, b] = [b, a] = [-a, b] = [a, -b] = [-a, -b]$
2. $[a^n, b^n] = [a, b]^n$
3. $[ka, kb] = |k|[a, b]$
4. $[a, b] = |ab|/(a, b)$

$$[a, b] = \frac{|ab|}{(a, b)}$$

$$d = (a, b) \quad d|a \rightarrow a = a'd$$

$$d|b \rightarrow b = b'd$$

$$(a', b') = 1 \rightarrow ab = a'b'd^2$$

$$\underline{[a', b']} = a'b' \quad a'b' = \frac{ab}{d^2}$$

$$[a, b] = [a'd, b'd] = d[a', b'] = da'b'$$

$$= \frac{ab}{d} = \frac{ab}{(a, b)} \quad \square$$

Some Problems

Problem: $[a, b, c] = \frac{abc}{(ab, ac, bc)}$

Solution:

$$\underline{[a, b, c]} = \underline{[[a, b], c]} = \left[\underline{\frac{ab}{(a, b)}}, c \right] = \frac{\frac{abc}{(a, b)}}{\left(\frac{ab}{(a, b)}, c \right)}$$

$$= \frac{\frac{abc}{(a, b)}}{\frac{(ab, ac, bc)}{(a, b)}} = \frac{abc}{(ab, ac, bc)}$$

$$\frac{1}{(a, b)} (ab, c \mid (a, b)) = \frac{1}{(a, b)} (ab, (ac, cb))$$

Division Algorithm

Theorem:

$$\forall a, b \neq 0 \exists q, r: \underline{a = bq + r}, 0 \leq r < |b|$$

Proof:

- For simplicity assume $a, b > 0$
- Consider $R = \{a - bq \mid a - bq \geq 0\}$
- R has a least element; called it r ; $r = a - bq$ for some q
- r must be smaller than b , otherwise
- $0 \leq r - b = a - bq - b = a - b(q + 1) \rightarrow r - b \in R$

$(a, b) \xleftarrow{\quad} (b, r) = d$
 $\xrightarrow{\quad} a > b > 0$
 $a = bq + r$
 $0 \leq r < b$
 $d \mid a$
 $d \mid b$
 $d \mid a - bq = r$
 $d \mid b$
 $d \mid r$
 $d \mid bq + r = a$
 $\Rightarrow (a, b) = d$
 عكس

Euclidean Algorithm

Assume $a, b > 0$, $r_0 = a, r_1 = b$

- $r_0 = r_1 q_0 + r_2, 0 < r_2 < r_1$
- $r_1 = r_2 q_1 + r_3, 0 < r_3 < r_2$
- $r_2 = r_3 q_2 + r_4, 0 < r_4 < r_3$
- ...
- $r_n = r_{n+1} q_n + r_{n+2}, 0 < r_{n+2} < r_{n+1}$
- $r_{n+1} = r_{n+2} q_{n+1}$

Then $(a, b) = (r_0, r_1) = (r_1, r_2) = \cdots = (r_n, r_{n+1}) = (r_{n+1}, r_{n+2}) = r_{n+2}$

Some Problems

Any number can be written in any of the following format

- $2k, 2k + 1$
- $3k, 3k + 1, 3k + 2$
- $4k, 4k + 1, 4k + 2, 4k + 3,$
- ... $5k, 5k + 1, 5k + 2$

$$n(n^4 - 1) = n(\underline{n^2 - 1})(n^2 + 1) = \\ = n(n-1)(n+1)(n^2 + 1)$$

Problem: show $120 | n^5 - n$ for odd n

$$n = 5k \pm 1 \\ \xrightarrow{5} n(n-1)(n+1)$$

Solution: $3 \times 5 \times 8 | n(n-1)(n+1)(n^2 + 1)$

$$n^2 = (25k^2 \pm 20k + 4)$$

We know $3 | n(n-1)(n+1)$

$$\underline{5 \nmid n(n-1)(n+1)} \rightarrow n = 5k \mp 2 \rightarrow 5 | \underline{n^2 + 1}$$

$$\underline{n = 2k + 1} \rightarrow 8 | \underbrace{(n-1)}_{2 \times 4} \underbrace{(n+1)}_1 (n^2 + 1)$$

$$4k + 2 = n - 1 \\ 4k + 2 + 2 = n + 1 = 4k'$$

Representation of Integers

Let $b > 1$. Any positive integer n can be written in form of $n = a_k b^k + \dots + a_1 b + a_0 = (a_k \dots a_1 a_0)_b$ where $0 \leq a_i < b$, a_i is called a digit in base b

Examples:

$$859 = 8 \times 10^2 + 5 \times 10 + 9$$

$$(10110)_2 = 1 \times 2^4 + 1 \times 2^2 + 1 \times 2 = (22)_{10}$$

$$(3A0F)_{16} = 3 \times 16^3 + 10 \times 16^2 + 15 \times 16 = (14863)_{10}$$

How to compute digits of n base b :

- simply apply division algorithm
- $n = bq_0 + a_0, q_0 = bq_1 + a_1, \dots$

Prime Numbers

Definition:

Any number greater than 1 whose factors are only 1 and itself is called a prime number. Otherwise; it is called composite.

عدد اول

Properties:

1. $p|ab \rightarrow p|a \vee p|b$
2. $\overline{(a,p)} = \overline{1} \vee \overline{p}$
3. Any number has a prime factor
4. Any composite n has a prime factor p s.t. $p \leq \sqrt{n}$
5. #primes is infinity
6. #primes in form of $ak+b$ where $(a,b)=1$ is infinity
7. $n = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_k^{\alpha_k}, \alpha_i \geq 1$
8. At least a prime number exists between n and $2n$

Some Proofs

Proof (3)

using strong induction

استقرا قوی

If n is prime, we are done. Otherwise $n = ab$ where $a, b > 1$. Now consider the prime factor of a which is a factor of n

Proof (4)

$n = ab$ where $a, b > 1$. then $\min(a, b) \leq \sqrt{n}$.

Now consider a prime factor of $\min(a, b)$

زمن خلوت Proof (5)

Assume all prime numbers are $\{p_1, \dots, p_k\}$

Consider $N = p_1 \dots p_k + 1$ which has a prime factor p .

$p|N$ $\rightarrow$ $(p, p_i) = 1$ $\rightarrow$ p is a new prime number.

Some Problems

Problem: find $\underline{n}$ s.t. $\underline{n} \nmid \underline{(n-1)!}$

Solution:

- If n is in form of $\underline{n = ab}$, $\underline{a, b > 1}$, $\underline{a \neq b}$, then $\underline{n \mid (n-1)!}$.
Otherwise $\underline{n = p}$ or $\underline{p^2}$ where p is prime
- $\underline{n = p}$ is of course is answer. If $\underline{n = p^2}$, number $p, 2p$ exist in $\underline{(n-1)!}$ for $\underline{p > 2}$. Just check $\underline{n = 2^2}$.

$2, p$ (اعداد اول)

Problem: find n s.t. $n^2 \nmid (n-1)!$

$$n^2 \nmid (n^2 - 1)!$$

$$2^2 \nmid (2^2 - 1)! \quad \checkmark \quad n = 2$$

Some Problems

Theorem:

The power of p in $n! = \sum_{i=1}^{\infty} \left\lfloor \frac{n}{p^i} \right\rfloor$

Problem: show $\frac{(n+m)!}{n!m!}$ is integer

Solution:

We have to show for any p , the power of p in $(n+m)!$ is at least the power of p in $n!m!$

It is sufficient to show $\left\lfloor \frac{n+m}{p^i} \right\rfloor \geq \left\lfloor \frac{n}{p^i} \right\rfloor + \left\lfloor \frac{m}{p^i} \right\rfloor$ for any i

Problem: show $k!^k \mid k^3!$

Solution: show

$$\binom{k}{p^i} \left\lfloor \frac{k}{p^i} \right\rfloor \leq \left\lfloor \frac{k^3}{p^i} \right\rfloor \quad \text{for any prime } p$$

$$n = 8$$

$$1 \times 2 \times 3 \times 4 \times 5 \times 6 \times 7 \times 8 = 8!$$

$$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow \downarrow \downarrow$$

$$\left\lfloor \frac{8}{2} \right\rfloor + \left\lfloor \frac{8}{2^2} \right\rfloor + \left\lfloor \frac{8}{2^3} \right\rfloor$$

$$4 + 2 + 1 = 7$$

$$\lfloor x+y \rfloor \geq \lfloor x \rfloor + \lfloor y \rfloor$$

Some Problems

Problem:

At least a prime number exists between n and $n!$

Solution:

$$k \notin \{2 \dots n\} \quad k | (n! - 1) \Rightarrow k \nmid n!$$

One way is to show $n! \geq 2n$. The other way is to look at $N = n! - 1$ which is relatively prime to any number equal or less than n . So, this has a prime factor which is greater than n and of course less than $n!$. $p | N$

$$n < p < n! \quad \square$$

Problem:

if p and $p^2 + 2$ are prime, then $p^3 + 2$ is prime $p = 3k + 1$

Solution:

$$p^2 + 2 = 9k^2 + 6k + 1 + 2 = 9k^2 + 6k + 3 = 3(3k^2 + 2k + 1)$$

$p = 3$ is the answer. Other prime numbers are of form $3(3k^2 + 2k + 1)$
 $3k + 1$ or $3k + 2$. For both $3 | p^2 + 2$ $p = 3k + 2 \Rightarrow p^2 + 2 = 9k^2 + 12k + 4 + 2 = 9k^2 + 12k + 6 = 3(3k^2 + 4k + 2)$

Congruence

Definition:

$$\underline{a} \equiv \underline{b} \pmod{\underline{m}} \leftrightarrow \underline{m} \mid a - b$$

Properties:

1. $a \equiv a \pmod{m}$
2. $a \equiv b \pmod{m}, b \equiv c \pmod{m} \rightarrow a \equiv c \pmod{m}$
3. $a \equiv b \pmod{m} \leftrightarrow a + c \equiv b + c \pmod{m}$
4. $a \equiv b \pmod{m} \rightarrow \underline{ac} \equiv \underline{bc} \pmod{m}$
5. $\underline{ac} \equiv \underline{bc} \pmod{m} \rightarrow a \equiv b \pmod{m/(m, c)}$
6. $a \equiv \underline{b} \pmod{\underline{m}} \rightarrow a^n \equiv b^n \pmod{m}$
7. $a \equiv b \pmod{m} \rightarrow P(a) \equiv P(b)$ where P is a polynomial
8. $a \equiv r \pmod{m}$ where $a = mq + r, 0 \leq r < m$
9. $0 \leq i \neq j < \underline{m} \rightarrow [i] \cap [j] = \emptyset$ where $\underline{[i]} = \{x \mid x \equiv i \pmod{m}\}$

Some Proofs

Proof (5)

$$\begin{aligned}
 (c, m) = d &\rightarrow \underline{c} = \underline{c'd}, \underline{m} = \underline{m'd}, (c', m') = 1 \\
 \underline{ac} \equiv \underline{bc} \pmod{\underline{m}} &\rightarrow \underline{m} \mid \underline{c(a-b)} \rightarrow \underline{m'd} \mid \underline{c'd(a-b)} \\
 &\rightarrow \underline{m'} \mid \underline{c'(a-b)}, (m', c') = 1 \rightarrow \underline{m'} \mid \underline{a-b} \rightarrow \underline{a \equiv b \pmod{m'}} \\
 &\quad \quad \quad \underline{m} = \underline{m'} \\
 &\quad \quad \quad (c, m) = d
 \end{aligned}$$

Proof (7)

$$\begin{aligned}
 \text{Let } P(x) &= \underline{p_k x^k} + \dots + \underline{p_1 x} + \underline{p_0} \\
 a \equiv b \pmod{m} &\rightarrow \forall i: \underline{p_i a^i} \equiv \underline{p_i b^i} \pmod{m} \rightarrow \underline{P(a)} \equiv \underline{P(b)} \pmod{m}
 \end{aligned}$$

Proof (9)

$$\begin{aligned}
 x \in [i] \cap [j] &\rightarrow x \equiv i \equiv j \pmod{m} \rightarrow m \mid i - j, 0 \leq i, j < m \\
 &\rightarrow i = j \\
 &\quad \quad \quad x \overset{m}{\equiv} i \quad x \overset{m}{\equiv} j \Rightarrow i \overset{m}{\equiv} j \quad i - j \cdot \cancel{X}
 \end{aligned}$$

Some Problems

فرد. زوج

Problem: $x \equiv 1 \pmod{2} \rightarrow x^2 \equiv 1 \pmod{8}$

Solution: $x \equiv 1 \pmod{2} \rightarrow x = 4k + 1 \vee 4k + 3 \rightarrow x^2 \equiv 1 \pmod{8}$
 $(4k+1)^2 = 16k^2 + 8k + 1$ $(4k+3)^2 = 16k^2 + 24k + 9$

Problem: Compute the remainder of 3×2^{1399} to 7

Solution:

$$2^3 \equiv 1 \pmod{7} \rightarrow 2^{3 \times 466} \equiv 1 \pmod{7} \rightarrow 2^{1398} \equiv 1 \pmod{7}$$

$$2^{1399} \equiv 2 \pmod{7} \rightarrow 3 \times 2^{1399} \equiv 6 \pmod{7} \text{ or simply write}$$

$$3 \times 2^{1399} \equiv 3 \times 2 \times 2^{3 \times 466} \equiv 6 \times 1^{466} \equiv 6 \pmod{7}$$

Some Problems

Problem:

Compute the rightmost digit of 1398^{1399} base 10

Solution:

$$\begin{aligned} 1398^{1399} &\equiv 8^{1399} \equiv (-2)^{1399} \equiv -2^{1399} \pmod{10} \\ -2^{1399} &\equiv -2^{4 \times 349 + 3} \end{aligned}$$

$$\begin{aligned} &(-2)^{1399} \\ -2 &= (-2)^1 \equiv 8 \\ -8 &= (-2)^2 \equiv 4 \\ &= (-2)^3 \equiv 2 \\ &= (-2)^4 \equiv 6 \\ &= (-2)^5 \equiv 8 \end{aligned} \left. \begin{array}{l} \\ \\ \\ \\ \end{array} \right\} \begin{array}{l} 1399 \equiv 3 \\ \Rightarrow (-2)^{1399} \equiv 2 \end{array}$$

$$-2^{1399} \equiv 2 \pmod{10} \rightarrow 1398^{1399} \equiv 2 \pmod{10}$$

Some Problems

Problem: $n = (a_k \dots a_0)_{10} \rightarrow n \equiv \underline{a_k + \dots + a_0} \pmod{9}$

Solution: $n = a_k \underbrace{10^k}_1 + \dots + a_1 10 + a_0, 10 \equiv 1 \pmod{9}$

Problem: Find all prime p and q s.t. $p^2 + 2q^2 = x^2$

Solution: $2q^2 = x^2 - p^2 = (x-p)(x+p) \rightarrow$

$p = 2 \rightarrow x = 2k \rightarrow q = 2$ but $(2,2)$ is not the answer

Otherwise, $\overline{p} = 2k + 1 \rightarrow \overline{x} = 2k' + 1 \rightarrow p^2 \equiv x^2 \equiv 1 \pmod{8}, \underline{p^2 + 2q^2} \equiv \underline{x^2} \pmod{8} \rightarrow \underline{2q^2} \equiv 0 \pmod{8} \rightarrow$

$\boxed{q = 2} \rightarrow (x-p)(x+p) = 8 \rightarrow$

$(x-p = 1 \wedge x+p = 8) \vee \underline{(x-p = 2 \wedge x+p = 4)} \rightarrow$

$x = 3, p = 1$ But 1 is not prime

$$\begin{aligned} 2q^2 &\equiv 8 \\ q^2 &\equiv 4 \\ q &\equiv 2 \end{aligned}$$

p, q (prime)

$$x^2 = 2q^2 + p^2$$

$p = 2, q \text{ odd}$

$$x^2 = 2q^2 + 4$$

$$\underbrace{11 \equiv 3}_{\neq} \underbrace{2 \times 1 + 0}_{(\text{mod } 4)}$$

$$q, x \rightarrow \begin{cases} 4k + 0/2 \rightarrow q \\ 4k + 1/3 \rightarrow x \end{cases}$$

$$(1j)x^2 = (4k \pm 1)^2 \equiv 1 \pmod{4}$$

$$(2j)x^2 = (4k + 0/2) \equiv 0 \pmod{4}$$

Euler's Totient Function

Definition:

$$\Phi(n) = \{x | (x, n) = 1, 1 \leq x \leq n.\}, \varphi(n) = |\Phi(n)|$$

$$\varphi(1) = 1, \varphi(2) = 1, \varphi(3) = 2, \varphi(4) = 2, \varphi(p) = p - 1$$

$$\text{Theorem: } n = p_1^{\alpha_1} \dots p_k^{\alpha_k} \rightarrow \varphi(n) = n \left(1 - \frac{1}{p_1}\right) \dots \left(1 - \frac{1}{p_k}\right)$$

Proof: show that $(m, n) = 1 \rightarrow \varphi(mn) = \varphi(m)\varphi(n)$

1 Lemma: $(a, n) = 1, i \neq j \in \Phi(n) \rightarrow ai \not\equiv aj \pmod{n}$

2 Lemma: $(a, n) = 1, i \in \Phi(n) \rightarrow \exists j \in \Phi(n): ai \equiv j \pmod{n}$

Then, there is a one-to-one correspondence between

$\Phi(n)$ and $\{ax | x \in \Phi(n)\} \pmod{n}$. Therefore,

$$\prod_{i \in \Phi(n)} i \equiv \prod_{i \in \Phi(n)} ai \pmod{n} \text{ and } \left(\prod_{i \in \Phi(n)} i, n\right) = 1 \rightarrow$$

$$\text{Theorem: } (a, n) = 1 \rightarrow a^{\varphi(n)} \equiv 1 \pmod{n}$$

$$\varphi(n) = n \left(1 - \frac{1}{p_1}\right) \left(1 - \frac{1}{p_2}\right) \dots \left(1 - \frac{1}{p_k}\right)$$

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_k^{\alpha_k}$$

$$= \underbrace{p_1^{\alpha_1} p_2^{\alpha_2} \dots p_k^{\alpha_k}}_n \left(1 - \frac{1}{p_1}\right) \left(1 - \frac{1}{p_2}\right) \dots$$

$$\varphi(nm) = \varphi(n) \varphi(m)$$

$$(n, m) = 1$$

$$\varphi(p_1^{\alpha_1}) = p_1^{\alpha_1} - p_1^{\alpha_1-1} = p_1^{\alpha_1} \left(1 - \frac{1}{p_1}\right)$$

$$1 \dots p_1^{\alpha_1}$$

$$1 \dots 9$$

$$p^{\alpha} = 3^2$$

$$\frac{p_1^{\alpha_1}}{p_1} = p_1^{\alpha_1-1} \rightarrow \text{غیر قابل}$$

$$3^2 - 3^{2-1} = 6 = \varphi(9)$$

$$\{3, 6, 9\} \rightarrow \{1, 2, 4, 5, 7, 8\}$$

غیر قابل

$$1/ \vdash : \left. \begin{array}{l} (a, n) = 1 \\ i, j \in \phi(n) \\ i \neq j \end{array} \right\} \Rightarrow a_i \not\equiv a_j$$

اثبات:

$$a_i \equiv a_j \quad : \text{ فرض خلاف}$$

$$\begin{array}{c} \downarrow \\ \underline{n} \mid \underline{a(i-j)} \xrightarrow{(a,n)=1} n \mid (i-j) \\ \downarrow \end{array}$$

$$\underline{\underline{\nexists i=j}} \leftarrow i-j=0 \leftarrow i, j \in [1, n] \rightarrow \underline{i-j} < n$$

$$2/ \vdash : \left. \begin{array}{l} (a, n) = 1 \\ i \in \phi(n) \end{array} \right\} \Rightarrow \exists j \in \phi(n), a_i \equiv j$$

اثبات:

$$\begin{array}{l} (a, n) = 1 \\ (n, i) = 1 \end{array} \rightarrow \underline{(a_i, n) = 1}$$

$$a_i = qn + r \quad \cdot \quad \cancel{r} < n$$

$$\rightarrow (n, r) = 1, r \neq 0 \Rightarrow r = j$$

$$(n \nmid a_i \uparrow j) \quad a_i \equiv r \quad \square$$

$$(a, n) = 1$$

$$\phi(n) = \{r_1, r_2, \dots, r_{\phi(n)}\}$$

$$ar_i \equiv r_{\sigma(i)} \quad (\text{طبق م ۲})$$

طبق م ۱، تایید یکتا است.

$$\prod_{i=1}^{\phi(n)} (ar_i) \equiv \prod_{i=1}^{\phi(n)} r_{\sigma(i)} \equiv \prod_{i=1}^{\phi(n)} r_i$$

$$\begin{array}{ccc} \downarrow & & \\ a^{\phi(n)} \prod_{i=1}^{\phi(n)} r_i & \stackrel{n}{\equiv} & \prod_{i=1}^{\phi(n)} r_i \end{array} \quad \begin{array}{c} (n, p) = 1 \\ \xrightarrow{p \nmid n} \\ p \end{array}$$

$$a^{\phi(n)} \equiv_n 1 \quad \square$$

* در اینجا کار را از r نشانه شده است.

Some Problems

Problem: if d is the smallest natural number s.t. $a^d \equiv 1 \pmod{n}$ and $a^m \equiv 1 \pmod{n}$, then $\underline{d} | \underline{m}$

Solution:

$$m = \underline{d}q + \underline{r}, 0 \leq r < d, a^m \equiv a^d \equiv 1 \pmod{n} \rightarrow$$

$$\underline{a}^r \equiv 1 \pmod{n} \rightarrow r = 0 \rightarrow d | m \quad \underline{a}^r = a^{m-dq} \equiv a^m \cdot (a^d)^{-q} \equiv 1 \cdot 1^{-q} \equiv 1$$

Problem: $n | \varphi(2^n - 1)$

Solution: $n < k \leftarrow 2^n - 1 \mid 2^k - 1 \leftarrow \boxed{2^k \equiv 1} \rightarrow$
 $\underline{n}$ is the smallest number s.t. $\underline{2^n \equiv 1} \pmod{\underline{2^n - 1}}$

Since $\underline{2^{\varphi(2^n - 1)}} \equiv 1 \pmod{2^n - 1}$, then $n | \varphi(2^n - 1)$

$$(2, 2^n - 1) = 1 \rightarrow$$

Wilson's Theorem

Definition:

a^* is called inverse of $a \pmod n$ iff $aa^* \equiv 1 \pmod n$

Lemma: (a^* exists iff $(a, n) = 1$) and $a^* \equiv a^{\varphi(n)-1} \pmod n$

Theorem: if p is prime, then $(p-1)! \equiv -1 \pmod p$

Proof:

- For any $a \in \{1, \dots, p-1\}$, inverse exists. $a^* a \equiv 1$
- If $a^* = a \rightarrow a^2 \equiv 1 \pmod p \rightarrow p \mid (a-1)(a+1) \rightarrow$
 $\underline{a = 1} \vee \underline{a = p-1}$ $(a, p) = 1 \rightarrow ax + py = 1$
- For other a , we have $a^* \neq a$
 $\cdot < r < p$
 $x = pq + r$
 $ax = a(pq+r) = apq + ar$
 $ax \equiv 1$
 $ar \equiv ax \equiv 1$
- Set $\{2, 3, \dots, p-2\}$ can be decomposed into disjoint
 $\leftarrow$ pairs (a, b) ($a \neq b$) s.t. $ab \equiv 1 \pmod p$
 $a \neq \frac{1}{a^*}$

Chinese Remainder Theorem

Theorem:

$\forall a, b \forall m, n \text{ s.t. } (m, n) = 1 \text{ we have}$

$$\exists x \left(\underline{x \equiv a \pmod{n}} \wedge \underline{x \equiv b \pmod{m}} \right)$$

There is an unique x in $[0, \underline{mn} - 1]$ satisfying above

Solution: $x \equiv \underline{bn^*n} + \underline{am^*m} \pmod{mn}$ where

$nn^* \equiv 1 \pmod{m}$ and $mm^* \equiv 1 \pmod{n}$

$$\begin{aligned} a &\stackrel{m}{\equiv} b \\ a &\stackrel{n}{\equiv} b \Rightarrow a \stackrel{nm}{\equiv} b \end{aligned} \quad (m, n) = 1$$

Theorem can be extended to k linear equations:

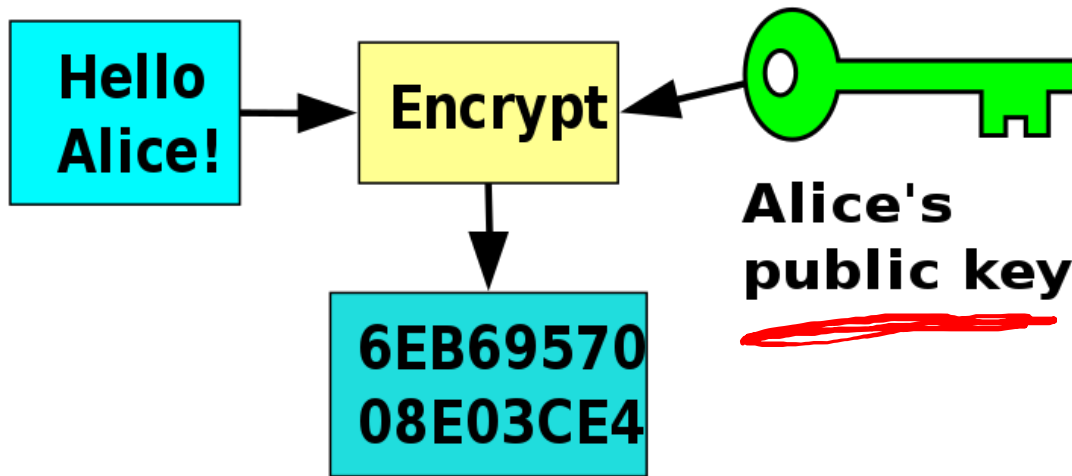
$x \equiv a_1 \pmod{n_1} \dots x \equiv a_k \pmod{n_k}$ where $(n_i, n_j) = 1$

Example: $7u + 9v = 1$ $u = 4$ $v = -3$ $x = a \cdot 9v + b \cdot 7u = 59$

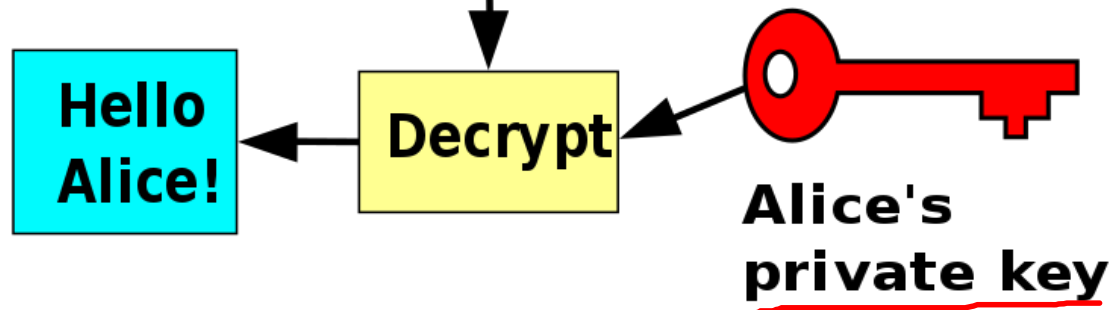
$$x \equiv \underline{3} \pmod{\underline{7}} \wedge x \equiv \underline{5} \pmod{\underline{9}} \rightarrow x \equiv 59 \pmod{\underline{63}}$$

Public Key Encryption

Bob



Alice



Public Key Encryption

Let $n = pq$ where p and q are large prime numbers

Bob's key is the pair n and e where e is the number s.t.

$(e, (p-1)(q-1)) = 1$. Anybody else may have this key (indeed it is a public key).

Alice's key is the pair p and q and a number d s.t. $de \equiv 1$
($\text{mod } (p-1)(q-1)$) (the key is private)

See message M as an integer number $(M, pq) = 1$ $M^{\varphi(pq)} \equiv 1$

Bob sends $C \equiv M^e \pmod{n}$ instead of sending M $\xrightarrow{\varphi(pq)}$

Alice computes $C^d \equiv M^{de} \equiv M^{k(p-1)(q-1)+1} \equiv M \pmod{n}$ To uniquely decrypt M , we need $M < n$. Then decompose original message into smaller pieces; each smaller than n

Without knowing p and q is hard to decrypt $M^e \pmod{n}$

It is hard (time-consuming) to decompose n to pq .